

# ERBOT Results Summary

Affiliation Strings Dataset

ERBOT v0.2.0

Generated: 2026-03-22 | Run time:  $\approx$ 5.6 hours

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## 1 Dataset Overview

| Property     | Value                                                              |
|--------------|--------------------------------------------------------------------|
| Dataset      | Affiliation Strings                                                |
| Records      | 2,260                                                              |
| Fields       | 2 ( <b>id</b> , <b>affiliation</b> )                               |
| Task         | Deduplication (single-source)                                      |
| Ground truth | 32,816 matched pairs (from <b>affiliationstrings_mapping.csv</b> ) |

Affiliation strings is a benchmark of research paper author affiliations — free-text strings such as “Dept. of Computer Science, MIT, Cambridge, MA” that refer to real-world institutions. The same institution appears under many surface forms: abbreviations, missing department names, country variants, spelling differences, and language variations. The dataset has only one content field (**affiliation**), making it a hard single-field deduplication task.

| Stage      | Setting         | Value                                            |
|------------|-----------------|--------------------------------------------------|
| Blocking   | Method          | Prefix blocking on <b>affiliation</b> , length 3 |
| Blocking   | Candidate pairs | 164,719                                          |
| Blocking   | Reduction ratio | 0.935 (93.5% of all pairs eliminated)            |
| Similarity | Fields computed | 1 ( <b>affiliation</b> only)                     |
| Weights    | Method          | Equal (weight = 1.0 for <b>affiliation</b> )     |
| Clustering | Methods run     | 9                                                |
| Merge      | Strategy        | Best                                             |

## 2 Pipeline Configuration

**Blocking note:** 164,719 pairs from 2,260 records (2,547,670 total possible). Prefix-3 on **affiliation** eliminates 93.5% of pairs. However, affiliation strings are highly variable in their opening tokens — “University of X”, “Dept. of Y, University of X”, “X University”, and “X Univ.” all refer to the same institution but have different 3-character prefixes. This means blocking likely discards many true matches before evaluation even begins, directly limiting recall for all methods.

## 3 Performance Results

All 9 clustering methods evaluated against the gold standard:

| Method              | ARI          | NMI          | VI          | B <sup>3</sup> -P | B <sup>3</sup> -R | B <sup>3</sup> -F | Hom.         | Com.  | V            |
|---------------------|--------------|--------------|-------------|-------------------|-------------------|-------------------|--------------|-------|--------------|
| <b>threshold_cc</b> | <b>0.076</b> | 0.688        | 4.24        | <b>0.395</b>      | 0.550             | <b>0.460</b>      | 0.606        | 0.796 | 0.688        |
| <b>louvain</b>      | <b>0.076</b> | 0.688        | 4.24        | 0.394             | 0.550             | 0.459             | 0.606        | 0.796 | 0.688        |
| <b>label_prop</b>   | <b>0.076</b> | 0.688        | 4.24        | 0.394             | 0.550             | 0.459             | 0.606        | 0.796 | 0.688        |
| leiden              | 0.000        | <b>0.819</b> | <b>3.42</b> | 1.000             | 0.146             | 0.255             | 1.000        | 0.693 | <b>0.819</b> |
| svm                 | 0.035        | <b>0.818</b> | <b>3.40</b> | <b>0.977</b>      | 0.159             | 0.273             | <b>0.990</b> | 0.697 | <b>0.818</b> |
| hclust_avg          | 0.053        | 0.516        | 5.84        | 0.184             | 0.530             | 0.272             | 0.404        | 0.716 | 0.516        |
| pam                 | 0.053        | 0.516        | 5.84        | 0.184             | 0.530             | 0.272             | 0.404        | 0.716 | 0.516        |
| hclust_ward         | 0.042        | 0.516        | 5.89        | 0.185             | 0.519             | 0.273             | 0.406        | 0.706 | 0.516        |
| gc                  | 0.000        | 0.000        | 7.72        | 0.007             | 1.000             | 0.014             | 0.000        | 1.000 | 0.000        |
| <i>final</i>        | 0.076        | 0.688        | 4.24        | 0.395             | 0.550             | 0.460             | 0.606        | 0.796 | 0.688        |

*Metrics: ARI = Adjusted Rand Index; NMI = Normalized Mutual Information; VI = Variation of Information (lower is better); B<sup>3</sup> = B-cubed; V = V-measure; Hom. = Homogeneity; Com. = Completeness. PairF columns in `er_performance.csv`.*

## 4 Final Cluster Assignment

The **final** strategy correctly selected **threshold\_cc** (highest ARI = 0.076).

| Statistic         | Value |
|-------------------|-------|
| Total clusters    | 290   |
| Records           | 2,260 |
| Mean cluster size | 7.8   |

290 clusters from 2,260 records is a reasonable partition for an affiliation dataset where each real institution may appear under dozens of surface forms.

## 5 Method-by-Method Interpretation

### Threshold-CC / Louvain / Label Propagation (Best — ARI = 0.076)

All three produce nearly identical results (as with CORA), again indicating the similarity graph has dominant structure that overwhelms algorithm differences. B<sup>3</sup>-F of 0.460 reflects a reasonable balance: precision 0.395 (some false merges) and recall 0.550 (misses about half of true matches). These graph-based methods are the practical ceiling given equal-weight single-field similarity.

### SVM (ARI = 0.035 — paradoxically worse than threshold-CC)

SVM has near-perfect precision (0.977) and near-perfect homogeneity (0.990) but extremely low recall (0.159). It is so conservative that it only merges pairs it is almost certain about, producing many small, pure clusters — but missing the vast majority of true duplicates. On a single-field dataset with high surface-form variability, SVM cannot generalise beyond the high-confidence pairs in its training signal, making it less useful than the simpler threshold approach. High NMI (0.818) reflects the purity of its clusters, not their coverage.

### Leiden (Degenerate — ARI = 0.000)

Perfect homogeneity (1.0) and highest NMI (0.819) but near-zero recall (0.146). Like SVM, Leiden over-partitioned — the resolution parameter is too high for this dataset, creating hundreds of tiny, pure sub-clusters instead of merging variants of the same institution. The VI score (3.42) is the best of the non-SVM methods despite ARI = 0, reflecting high per-cluster purity. Needs a lower resolution parameter.

### GC — Graph Coloring (Degenerate — ARI = 0.000)

Perfect recall (1.0) but near-zero precision (0.007), collapsing all 2,260 records into effectively one cluster. GC continues to fail at the default threshold — the dimension mismatch error also caused it to fall back to a degenerate solution silently.

### PAM / hclust\_avg / hclust\_ward (ARI ≈ 0.042–0.053)

All three are below threshold-CC. They find some structure but the similarity matrix from a single noisy text field is insufficiently informative to guide centroidal or hierarchical methods toward the true partition.

## 6 Key Findings

1. **Overall performance is low.** Best ARI is 0.076 — far below CORA (0.746). This is expected: affiliation deduplication is intrinsically harder because (a) only one field is available, (b) surface-form variability is extreme (abbreviations, reorderings, language variants), and (c) prefix-3 blocking likely discards many true matches before evaluation.
2. **Blocking is the primary bottleneck.** Prefix-3 on **affiliation** is poorly suited to this dataset. Affiliation strings start with highly variable tokens (“University of”, “Dept.”, city names, acronyms). Many true duplicates will have non-matching 3-character prefixes and are silently dropped at Stage 3. This single issue likely accounts for most of the low recall across all methods.
3. **SVM precision–recall tradeoff is extreme.** Precision = 0.977, recall = 0.159. SVM learns to be extremely conservative on this noisy single-field data — it finds the easy matches

but ignores the hard ones. For high-precision applications SVM is useful; for high-recall applications it is not.

4. **Graph methods are the practical best.** Threshold-CC, Louvain, and Label Propagation achieve the best balanced B<sup>3</sup>-F (0.460) and are robust, consistent, and fast relative to SVM.
5. **Equal weights are not the issue here.** With only one field, weight learning cannot help. The limiting factors are blocking and similarity function choice.
6. **The final merge correctly picked `threshold_cc`.** The merge bug fix is working as expected across datasets.

## 7 Recommendations for Next Steps

| Priority | Action                                                                                                                                                                                  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High     | Replace prefix-3 blocking with SN blocking (sorted neighbourhood, window = 40–60) on <code>affiliation</code> — far better suited to variable-length strings with no consistent prefix. |
| High     | Use token-set ratio or Jaccard on token sets as the similarity function — affiliation strings are bag-of-words problems, not edit-distance problems.                                    |
| Medium   | Tune Leiden resolution (try 0.05–0.2) to reduce over-splitting.                                                                                                                         |
| Medium   | Fix GC dimension mismatch error — <code>gc</code> falls back to a degenerate solution silently.                                                                                         |
| Medium   | Measure pair completeness at Stage 3 to quantify how many true matches blocking discards.                                                                                               |
| Low      | Try TF-IDF + cosine similarity with IDF trained on the full affiliation corpus — common tokens like “University” and “Department” should be down-weighted.                              |
| Low      | Try record linkage with an external affiliation authority list (ROR, OpenAlex institutions) as a lookup layer before ER.                                                                |

## 8 Comparison with CORA

| Metric                 | CORA                    | Affiliation               | Interpretation                     |
|------------------------|-------------------------|---------------------------|------------------------------------|
| Records                | 1,879                   | 2,260                     | Similar scale                      |
| Fields                 | 15                      | 1                         | Affiliation severely disadvantaged |
| Best ARI               | 0.746                   | 0.076                     | ≈10× harder                        |
| Best B <sup>3</sup> -F | 0.738                   | 0.460                     | Large gap                          |
| Best method            | <code>hclust_avg</code> | <code>threshold_cc</code> | Different winners                  |
| Final correct          | Yes                     | Yes                       | Merge fix working                  |
| Blocking RR            | 92.2%                   | 93.5%                     | Similar reduction                  |
| Runtime                | ≈63 min                 | ≈338 min                  | SVM bottleneck                     |

The performance gap is almost entirely explained by the field count: CORA has 15 discriminative fields that jointly identify duplicates; affiliation has one highly variable text field with no supporting structure.

## 9 Output Files

| File                         | Description                                                  |
|------------------------------|--------------------------------------------------------------|
| er_performance.csv           | Full metric table (10 methods $\times$ 13 metrics)           |
| er_predictions.csv           | Final cluster assignments (2,260 records $\times$ 2 columns) |
| affiliation_data_profile.pdf | Dataset profile: field types, distributions, missingness     |
| plot_method_comparison.pdf   | Grouped bar chart: ARI, B <sup>3</sup> -F, NMI per method    |
| plot_performance_heatmap.pdf | Colour heatmap: all metrics $\times$ all methods             |
| plot_cluster_sizes.pdf       | Cluster size distribution histogram                          |
| plot_precision_recall.pdf    | Precision vs. Recall scatter (B-cubed and Pair-F)            |
| AFF_RESULTS_SUMMARY.pdf      | This document                                                |